



Data Engineering

SETUP GIT & GITHUB



What is Git?

🌱 Git is a time machine for code that lets you confidently break everything today and magically restore it tomorrow. 🤪

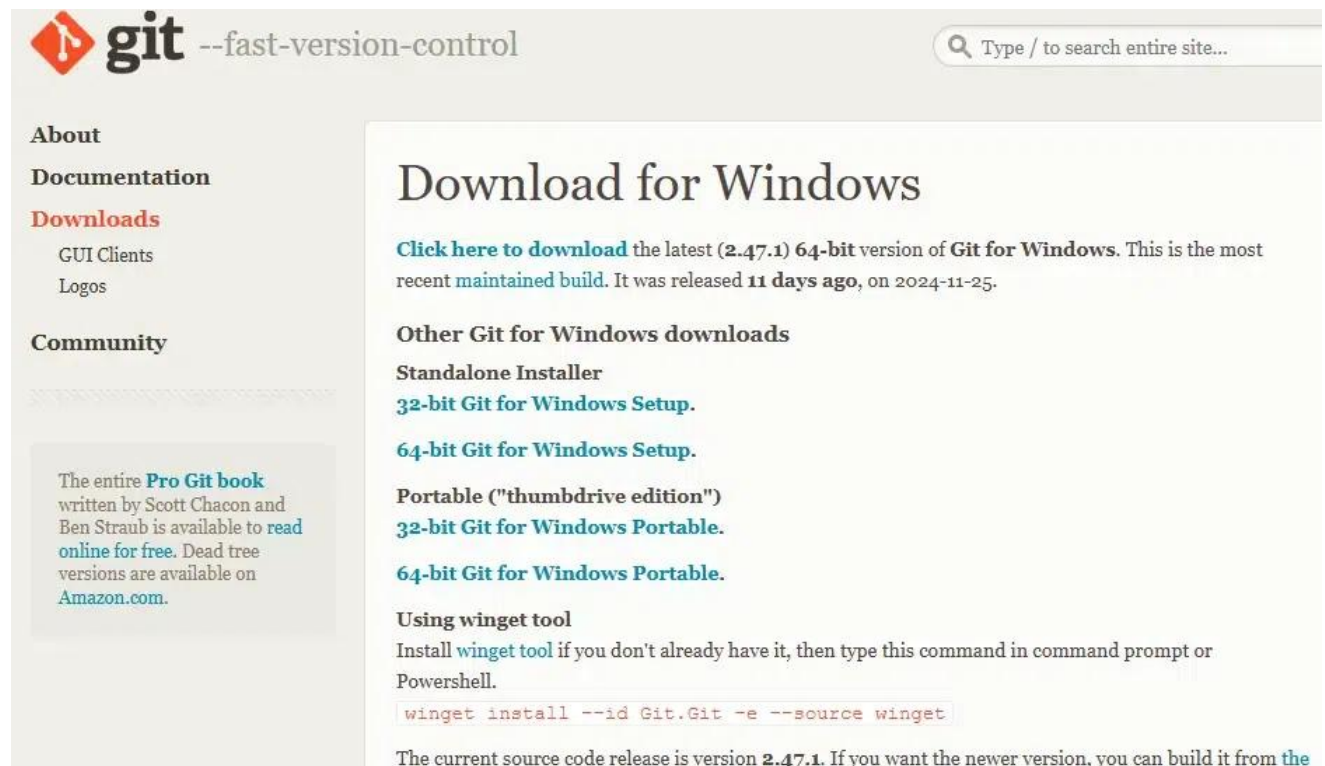
Git is a distributed version control system for tracking file changes, while GitHub is a cloud platform for hosting Git repositories and enabling collaboration. Installing Git is required to use both for version control.

Git on Windows

We will install git on Windows through the official Git website, which is the easy and most recommended way. The following steps are for installing the git on Windows.

Step 1: Download the Installer

Go to the official Git website: <https://git-scm.com/downloads/win>.



The screenshot shows the official Git website. The header includes the Git logo and the tagline "--fast-version-control". A search bar is located in the top right corner. The left sidebar contains navigation links: "About", "Documentation", "Downloads" (highlighted in red), "GUI Clients", "Logos", and "Community". The main content area is titled "Download for Windows". It features a link to download the latest (2.47.1) 64-bit version of Git for Windows, noting it's the most recent maintained build released 11 days ago. Below this, there are sections for "Other Git for Windows downloads" including "Standalone Installer" with links for 32-bit and 64-bit setups, and "Portable ('thumbdrive edition')" with links for 32-bit and 64-bit portable versions. A section titled "Using winget tool" provides instructions to install winget and a command to install Git: `winget install --id Git.Git -e --source winget`. At the bottom, it mentions the current source code release is version 2.47.1 and provides a link to build it from the source.

The download will start automatically for the latest version of Git for Windows. After the download is complete, run the .exe file.

Step 2: Select Editor & Adjust Path

Follow the prompts in the setup wizard. Most default settings will be fine for general use, but here are some key steps:

- Firstly, the installer will ask which text editor to use for [Git](#). You can choose from options like Vim, Notepad++, or Visual Studio Code.

- Make sure you choose the option that adds Git to your system PATH (recommended for ease of use in the command line).
- Select the default option (OpenSSL) to allow Git to communicate securely over [HTTPS](https://).
- The default choice, "Checkout Windows-style, commit Unix-style line endings," is ideal for most developers working in Windows environments.

Step 3: Complete the Installation

After configuring the setup, click "Install" and allow the installation to complete.

Once done, you can launch Git Bash (installed with Git) or use Git from the Command Prompt (cmd) by typing the following command.

```
git --version
```

```
C:\Users\GFG0536>git --version
git version 2.42.0.windows.1

C:\Users\GFG0536>|
```

If Git is installed, it will display the version number.

Git on Linux

Step 1: Update the System

For Debian/Ubuntu:

```
sudo apt update
```

```
sudo apt install git
```

For Fedora:

```
sudo dnf install git
```

For Arch Linux:

```
sudo pacman -S git
```

Step 2: Verify Installation

Use the below command to verify the installation of Git in Ubuntu.

git --version

```
adminiistrator@GFG19298-ANKIT-TESTING: ~  
Setting up libb-hooks-endofscope-perl (0.24-1) ...  
Setting up libnamespace-clean-perl (0.27-1) ...  
Setting up libnamespace-autoclean-perl (0.29-1) ...  
Setting up libdatetime-locale-perl (1:1.25-1) ...  
Setting up libdatetime-timezone-perl (1:2.38-1+2019c) ...  
Setting up libdatetime-perl:amd64 (2:1.51-1build1) ...  
Setting up libdatetime-format-strptime-perl (1.7600-1) ...  
Setting up libdatetime-format-builder-perl (0.8200-1) ...  
Setting up libdatetime-format-iso8601-perl (0.08-2) ...  
Setting up git-mediawiki (1:2.25.1-1ubuntu3.13) ...  
Setting up git-all (1:2.25.1-1ubuntu3.13) ...  
Processing triggers for fontconfig (2.13.1-2ubuntu3) ...  
Processing triggers for desktop-file-utils (0.24-1ubuntu3) ...  
Processing triggers for mime-support (3.64ubuntu1) ...  
Processing triggers for hicolor-icon-theme (0.17-2) ...  
Processing triggers for gnome-menus (3.36.0-1ubuntu1) ...  
Processing triggers for libc-bin (2.31-0ubuntu9.16) ...  
Processing triggers for ufw (0.36-6ubuntu1.1) ...  
Processing triggers for systemd (245.4-4ubuntu3.24) ...  
Processing triggers for man-db (2.9.1-1) ...  
Processing triggers for install-info (6.7.0.dfsg.2-5) ...  
adminiistrator@GFG19298-ANKIT-TESTING:~$ git --version  
git version 2.25.1  
adminiistrator@GFG19298-ANKIT-TESTING:~$
```

Git on Mac

Step 1: Get Homebrew in your macOS

If you don't have Homebrew, then type the following command in the terminal:

```
/bin/bash -c "$(curl -fsSL
```

```
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

```
gfg0458 -- -zsh -- 85x35
Last login: Thu Dec  5 16:15:56 on ttys000
gfg0458@GFG0458-VAIBHAV ~ % /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
-
```

Step 2: Install Brew

We recommend you install Homebrew first and then run the below command to download Git with no errors:

brew install git

```
gfg0458 -- -zsh -- 145x36
Last login: Thu Dec  5 16:15:56 on ttys000
gfg0458@GFG0458-VAIBHAV ~ % brew install git
```

Step 3: Verify the Installation

Once the installation is complete, verify the installation:

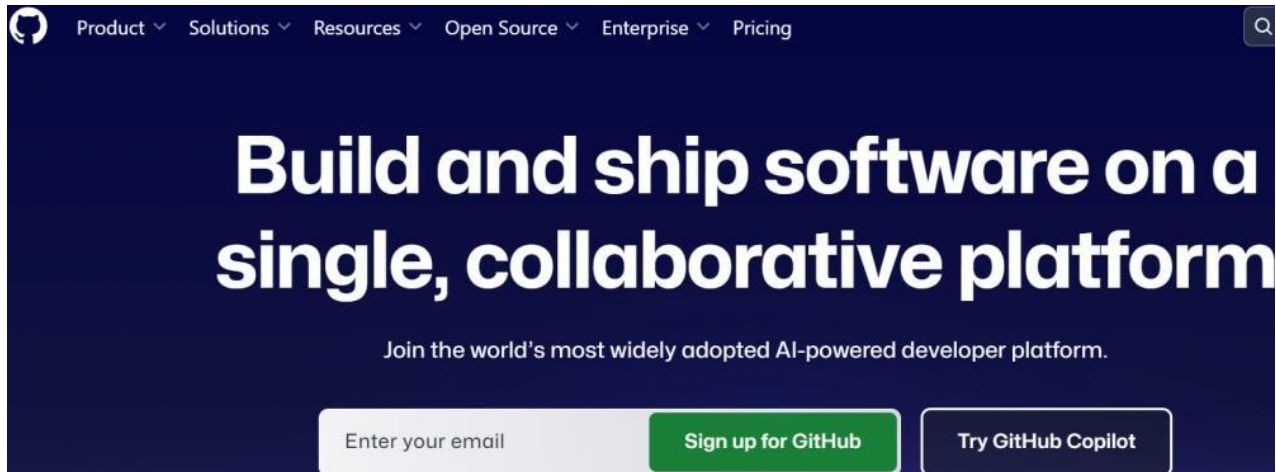
git --version

```
gfg0458 -- -zsh -- 145x36
Last login: Thu Dec  5 16:15:56 on ttys000
gfg0458@GFG0458-VAIBHAV ~ % git --version
```


Setting Up GitHub with Git

Step 1: Create a [GitHub](#) Account

- Visit GitHub and sign up.



Step 2: Configure Git with GitHub

Run the following commands in the terminal (Windows Git Bash, Linux, or Mac Terminal):

```
git config --global user.name "YourName"
```

```
git config --global user.email "youremail@example.com"
```

Step 3: Generate SSH Key (Optional but Recommended)

```
ssh-keygen -t rsa -b 4096 -C "youremail@example.com"
```

- Copy the key:

```
cat ~/.ssh/id_rsa.pub
```

- Add it to GitHub > Settings > SSH and GPG keys.

Step 4: Create a New Repository

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere?

[Import a repository.](#)

Required fields are marked with an asterisk ().*

Owner *

Repository name *

 GeeksforgeeksDS ▾ /

Great repository names are short and memorable. Need inspiration? How about **legendary-rotary-phone** ?

Description (optional)



Public

Anyone on the internet can see this repository. You choose who can commit.



Private

You choose who can see and commit to this repository.